

An Explainable Fraud Intelligence Framework for Operational Fraud Detection and Alert Prioritisation

Business Report

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Submission Date: 28 June 2026

Executive Summary

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Financial institutions operate in an increasingly complex fraud landscape where the growth of digital payment channels, higher transaction volumes and rapidly evolving fraud techniques place significant pressure on fraud operations. While machine learning has improved the ability to identify suspicious transactions, many predictive models provide limited transparency, making it difficult for investigators to understand, justify and prioritise fraud alerts within operational environments.

*This project presents an **Explainable Fraud Intelligence Framework** developed as a proof-of-concept for a JP Morgan-inspired fraud analytics simulation. The objective was not only to improve fraud detection, but to demonstrate how predictive analytics, explainability and operational decision support can be integrated into a single framework that assists fraud investigators in making transparent and informed decisions.*

The framework was developed using a synthetic banking transaction dataset representing customer, merchant and transaction activity. A structured analytical methodology was followed, including data preparation, exploratory data analysis, feature engineering, leakage auditing, class imbalance assessment, supervised machine learning, model validation and threshold evaluation. Two supervised learning algorithms—Logistic Regression and XGBoost—were evaluated to assess their suitability for operational fraud detection.

Model selection was based on both analytical performance and operational applicability. Logistic Regression was selected as the operational model because it achieved the strongest balance between fraud detection capability, interpretability and transparency, making it more suitable for investigator-led decision support. XGBoost was retained as a benchmark model and utilised for SHAP-based explainability to compare feature importance and reinforce model interpretation.

Beyond predictive modelling, the project developed an operational fraud intelligence framework incorporating risk categorisation, fraud alert prioritisation, reason codes, recommended investigator actions, service-level agreement (SLA) priorities and operational key performance indicators. These capabilities were integrated into interactive Power BI dashboards designed to support executive oversight, fraud monitoring and investigator workflow management.

The project also considers responsible AI principles by incorporating governance considerations, model documentation, explainability and operational limitations. Collectively, these components demonstrate how explainable machine learning can support transparent fraud operations while maintaining human oversight and improving decision-making.

Overall, the Explainable Fraud Intelligence Framework illustrates how data science, explainable artificial intelligence and business intelligence can be combined to deliver a practical decision-support solution for modern fraud operations. Although developed within a simulated banking environment, the framework provides a scalable foundation for future enhancements, including real-time fraud monitoring, advanced behavioural analytics and enterprise deployment.

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1. Introduction

1.1 Background

The rapid expansion of digital banking, electronic payments and online financial services has transformed the way financial institutions operate. While these advancements have improved customer accessibility and transaction efficiency, they have also created new opportunities for increasingly sophisticated fraudulent activity. Financial institutions must therefore detect suspicious transactions quickly while maintaining operational efficiency and minimising disruption to legitimate customer activity.

Machine learning has become a key enabler of modern fraud detection by identifying complex behavioural patterns that are difficult to capture using traditional rule-based systems alone. However, predictive performance is only one aspect of an effective fraud detection solution. Within operational environments, fraud investigators require transparent and explainable outputs that support informed decision-making, efficient alert prioritisation and effective case management.

This project addresses these challenges through the development of an **Explainable Fraud Intelligence Framework**. Rather than focusing solely on fraud prediction, the framework integrates machine learning, explainable artificial intelligence (XAI), operational risk prioritisation and business intelligence dashboards to support transparent, investigator-led fraud operations within a simulated banking environment.

1.2 Project Aim

This project aims to develop an Explainable Fraud Intelligence Framework that enhances operational fraud detection by combining supervised machine learning, explainability techniques and business intelligence into a transparent decision-support solution for fraud investigators.

1.3 Project Objectives

The project was undertaken to achieve the following objectives:

- Develop an operational fraud intelligence framework using a synthetic banking transaction dataset.
- Prepare and analyse customer, merchant and transaction data to identify fraud-related behavioural patterns.

- Engineer operational risk features to improve fraud detection and model interpretability.
- Develop and evaluate supervised machine learning models for fraud classification.
- Select an operational model that balances predictive performance with explainability.
- Incorporate explainability techniques to improve transparency and investigator confidence.
- Design an operational fraud prioritisation process using risk categories, reason codes, recommended investigator actions and service-level agreement (SLA) priorities.
- Develop interactive Power BI dashboards to support executive reporting and operational fraud investigation.
- Evaluate governance, responsible AI considerations and opportunities for future enhancement.

1.4 Report Structure

The remainder of this report presents the business problem addressed by the project, reviews the relevant literature, describes the methodology adopted to develop the framework and discusses the analytical findings. The report then examines the operational dashboards, business recommendations, governance considerations, project limitations and future enhancements before concluding with the overall outcomes of the Explainable Fraud Intelligence Framework.

2. Business Problem and Project Objectives

2.1 Business Context

Financial institutions operate within an increasingly complex fraud environment driven by the continued growth of digital banking, online payments and real-time financial transactions. While digital transformation has improved customer experience and operational efficiency, it has also expanded the attack surface available to fraudsters. As fraud techniques continue to evolve, organisations must detect suspicious activity quickly while minimising financial losses, protecting customers and maintaining regulatory compliance.

Traditional fraud detection approaches, which rely primarily on predefined business rules, remain effective for identifying known fraud scenarios but often struggle to detect emerging fraud patterns. These systems may also generate high volumes of false positive alerts, increasing investigator workload and reducing operational efficiency. Consequently, many financial institutions have adopted machine learning to improve fraud detection by identifying complex behavioural patterns within transactional data.

2.2 Problem Statement

Although machine learning has significantly improved fraud detection capability, many predictive models operate with limited transparency. Fraud investigators are often presented with a fraud score or classification without sufficient explanation of the underlying factors influencing the decision. This lack of interpretability can reduce investigator confidence, slow case resolution and create challenges for governance, auditability and responsible AI adoption.

The challenge addressed by this project is therefore not only to identify potentially fraudulent transactions but also to present fraud risk in a manner that is transparent, explainable and operationally actionable. The framework seeks to support investigators by combining predictive analytics with meaningful operational insights that enhance decision-making rather than replacing human judgement.

2.3 Project Scope

This project develops an Explainable Fraud Intelligence Framework within a simulated banking environment using a synthetic transaction dataset. The scope includes data preparation, exploratory analysis, feature engineering, supervised machine learning, explainability analysis and the development of interactive Power BI dashboards to support fraud investigation and operational reporting.

The project is designed as a proof-of-concept and does not represent a production-ready fraud detection platform. Real-time transaction processing, live system integration and automated fraud decision-making are outside the scope of this implementation. Human investigators remain responsible for reviewing prioritised alerts and making final operational decisions.

2.4 Success Criteria

The success of the framework was evaluated against the following objectives:

- Develop a transparent and explainable fraud intelligence framework suitable for operational decision support.
 - Select a machine learning model that balances fraud detection capability with interpretability.
 - Demonstrate explainability through operational risk indicators and model interpretation techniques.
 - Implement an operational alert prioritisation process using risk categories, reason codes, recommended investigator actions and SLA priorities.
 - Deliver interactive Power BI dashboards that support executive oversight and fraud investigation.
 - Incorporate governance and responsible AI considerations to promote transparent and accountable operational use.
 - Produce a professional proof-of-concept demonstrating how explainable analytics can strengthen modern fraud operations.
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3. Literature Review

3.1 Fraud Detection in Financial Services

The increasing adoption of digital banking and electronic payment systems has fundamentally changed the fraud landscape within financial services. Financial institutions now process millions of transactions daily across multiple channels, requiring fraud detection systems capable of identifying suspicious behaviour accurately and efficiently. Traditional rule-based systems remain valuable for detecting known fraud scenarios but are often limited when responding to rapidly evolving fraud techniques and previously unseen attack patterns. As a result, organisations are increasingly adopting data-driven analytical approaches that improve fraud detection while maintaining operational efficiency and customer trust.

3.2 Machine Learning for Fraud Detection

Machine learning has become a widely adopted approach for fraud detection because it can identify complex behavioural relationships within transactional data that are difficult to define using static business rules. Supervised learning algorithms have demonstrated strong performance in transaction classification by learning patterns associated with historical fraudulent activity. However, fraud detection datasets are typically characterised by class imbalance, requiring appropriate evaluation

using metrics such as precision, recall, ROC-AUC and confusion matrices rather than relying solely on overall accuracy. Consequently, effective model selection should consider both predictive performance and operational suitability.

3.3 Explainable Artificial Intelligence (XAI)

As machine learning models become increasingly sophisticated, explainability has emerged as an essential requirement within regulated industries such as banking. Explainable Artificial Intelligence (XAI) enables users to understand how model predictions are generated and which variables contribute most significantly to individual decisions. Greater transparency improves investigator confidence, supports governance requirements and facilitates responsible AI adoption by enabling fraud analysts to justify operational decisions using interpretable evidence rather than relying solely on prediction scores.

3.4 Operational Fraud Intelligence

Modern fraud operations require more than accurate prediction models. Effective fraud management combines predictive analytics with operational processes that enable investigators to prioritise alerts, allocate resources efficiently and make informed decisions. Risk categorisation, alert prioritisation, reason codes, recommended investigator actions and business intelligence dashboards provide operational context that improves workflow efficiency while supporting consistent fraud investigation across large transaction volumes.

3.5 Research Gap

Although existing research demonstrates the effectiveness of machine learning and explainable AI for fraud detection, many implementations focus primarily on improving predictive accuracy without integrating operational decision support into a single framework. This project addresses that gap by developing an Explainable Fraud Intelligence Framework that combines supervised machine learning, explainability techniques, operational fraud prioritisation and interactive business intelligence dashboards within one cohesive proof-of-concept. The resulting framework demonstrates how analytical models can support transparent, investigator-led fraud operations rather than functioning solely as predictive tools.

4. Project Methodology

4.1 Project Approach

This project followed a structured data analytics methodology to develop an Explainable Fraud Intelligence Framework for operational fraud detection within a simulated banking environment. The methodology combined data preparation, exploratory analysis, supervised machine learning, explainability techniques and business intelligence reporting to produce an end-to-end fraud intelligence solution. Rather than focusing solely on model development, the project emphasised transparency, operational decision support and practical applicability for fraud investigation.

4.2 Dataset and Data Preparation

The framework was developed using a synthetic banking transaction dataset comprising customer, merchant and transaction information. The individual datasets were integrated to create a unified analytical dataset suitable for fraud modelling and operational reporting.

Data preparation included validation of data quality, verification of data types, treatment of missing values where required and preparation of features for machine learning. A leakage audit was also performed to ensure that variables capable of revealing the target outcome or introducing future information were excluded from model development. This process helped ensure that model performance reflected genuine predictive capability rather than information leakage.

4.3 Feature Engineering

Feature engineering focused on creating variables that reflected operational fraud risk while improving model interpretability. Customer characteristics, merchant information and transactional attributes were combined with engineered behavioural indicators, including transaction timing, device familiarity, customer risk segmentation, KYC risk bands and merchant categorisation.

To support operational fraud investigation, the framework also incorporated explainable business features such as operational risk categories, fraud reason codes, recommended investigator actions and service-level agreement (SLA) priorities. These additions extended the framework beyond predictive modelling by supporting transparent fraud prioritisation and investigator workflow management.

4.4 Model Development and Evaluation

Two supervised machine learning algorithms, Logistic Regression and XGBoost, were developed and evaluated using a consistent modelling pipeline. The analytical workflow included train-test splitting, preprocessing, class imbalance handling, cross-validation and threshold evaluation to ensure reliable model assessment.

Model performance was evaluated using precision, recall, ROC-AUC and confusion matrices to balance fraud detection capability with operational practicality. Rather than selecting the model with the highest predictive performance alone, the evaluation considered explainability, transparency and suitability for operational deployment within an investigator-led fraud management process.

4.5 Explainability and Dashboard Integration

The final stage of the methodology focused on transforming analytical outputs into operational decision support. Explainability techniques were incorporated to improve transparency and enable investigators to understand the factors influencing fraud predictions. In parallel, interactive Power BI dashboards were developed to present fraud trends, operational KPIs, alert prioritisation and investigative insights in a format suitable for executive monitoring and operational decision-making.

The integration of predictive analytics, explainability and business intelligence formed the foundation of the Explainable Fraud Intelligence Framework, demonstrating how machine learning can support transparent and accountable fraud operations.

5. Framework Development and Results

5.1 Dataset Overview

The Explainable Fraud Intelligence Framework was developed using a synthetic banking transaction dataset representing customer, merchant and transaction activity within a simulated financial services environment. Customer, merchant and transaction datasets were integrated to create a unified fraud intelligence dataset that formed the foundation for both predictive modelling and business intelligence reporting.

The dataset contained transactional attributes, customer characteristics and merchant information together with engineered operational risk features developed during the analytical process. These variables enabled the framework to evaluate fraud risk from multiple perspectives while supporting operational explainability and investigator decision-making. As the dataset was synthetically

generated, no personally identifiable information was processed, making the project suitable for demonstrating fraud analytics techniques within a controlled environment.

	Metric	Value
0	Transaction Records	5000
1	Customer Records	5000
2	Merchant Records	5000
3	Fraud Cases	201
4	Legitimate Transactions	4799
5	Fraud Rate (%)	4.02
6	Modelling Grain	One transaction per row
7	Target Variable	fraud_label

Table 1. Summary of the analytical dataset used to develop the Explainable Fraud Intelligence Framework.

The above summarises the analytical dataset used to develop the Explainable Fraud Intelligence Framework. The low fraud rate reflects the class imbalance commonly observed in fraud detection datasets and informed the model evaluation strategy adopted throughout this project.

5.2 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was undertaken to understand transaction behaviour, identify fraud patterns and support feature engineering. The analysis examined fraud distribution across transaction channels, merchant categories, customer segments and behavioural characteristics to identify variables that contributed to elevated fraud risk.

The results indicated that fraud was not uniformly distributed across the dataset. Certain transaction channels, merchant categories and behavioural indicators demonstrated higher concentrations of fraudulent activity, highlighting the importance of combining multiple risk indicators rather than relying on individual variables. Additional analysis showed increased fraud occurrence for unfamiliar devices and transactions conducted during higher-risk periods, reinforcing the value of behavioural features within the fraud intelligence framework.

The findings from the exploratory analysis informed subsequent feature engineering and supported the development of operational risk categories used throughout the framework.

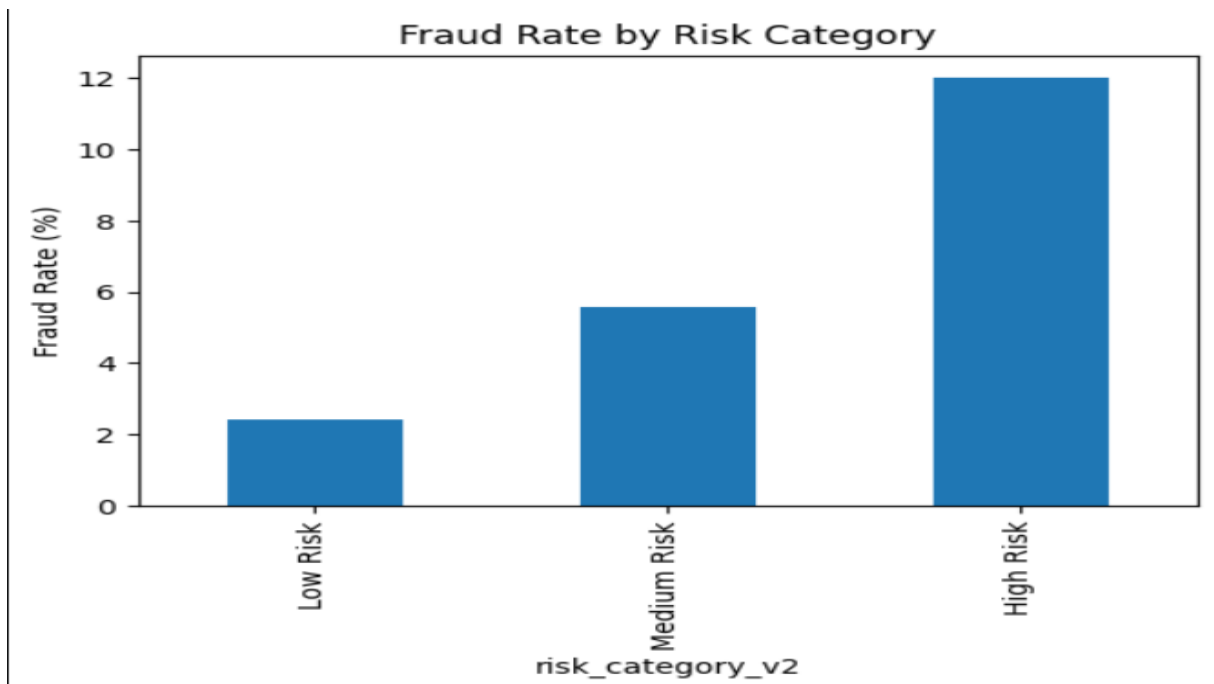


Figure 1. Fraud rates across operational risk categories demonstrate increasing fraud prevalence from Low-Risk to High-Risk transactions.

5.3 Model Performance

Two supervised machine learning algorithms, Logistic Regression and XGBoost, were evaluated using a consistent analytical pipeline. Model performance was assessed using precision, recall, ROC-AUC and confusion matrices to ensure that the evaluation considered both predictive capability and operational suitability.

Given the inherent class imbalance within fraud detection datasets, model selection was not based solely on overall accuracy. Greater emphasis was placed on identifying fraudulent transactions effectively while maintaining an acceptable balance between false positive alerts and investigator workload. Threshold evaluation further demonstrated how model performance could be adjusted to support different operational priorities depending on fraud investigation capacity and risk appetite.

Metric	Logistic Regression	XGBoost
ROC-AUC	0.628	0.595
Fraud Precision	0.08	0.099
Fraud Recall	0.525	0.25
Operational Explainability	High	Moderate
Operational Role	Selected Operational Model	Benchmark Model for SHAP

Table 2. Comparative performance of the Logistic Regression and XGBoost models evaluated within the Explainable Fraud Intelligence Framework.

5.4 Operational Model Selection

Comparative evaluation demonstrated that both models provided valuable analytical insights; however, they exhibited different operational characteristics. Logistic Regression delivered a stronger balance between fraud detection capability, transparency and interpretability, making it well suited for investigator-led fraud operations where understanding model decisions is essential.

XGBoost provided a valuable benchmark for comparative analysis and demonstrated the ability to capture more complex feature interactions. However, the reduced transparency associated with gradient boosting models limited its suitability as the primary operational model within this framework.

Consequently, Logistic Regression was selected as the operational model because it aligned more closely with the project's objective of developing an explainable fraud intelligence framework that supports transparent decision-making rather than maximising predictive performance alone.

5.5 Explainability Results

Explainability formed a central component of the framework by enabling fraud predictions to be interpreted within an operational context. SHAP analysis was applied to the benchmark XGBoost model to examine feature-level contributions, while the selected Logistic Regression model provided inherently interpretable outputs suitable for operational deployment.

Beyond model explainability, the framework translated analytical outputs into operational intelligence through risk categorisation, fraud reason codes, recommended investigator actions and SLA priorities. These explainable outputs enabled investigators to understand not only the predicted fraud risk but also the operational context supporting each prioritised alert.

Collectively, these capabilities transformed predictive modelling into an operational decision-support framework capable of improving transparency, investigator confidence and fraud prioritisation within a simulated banking environment.

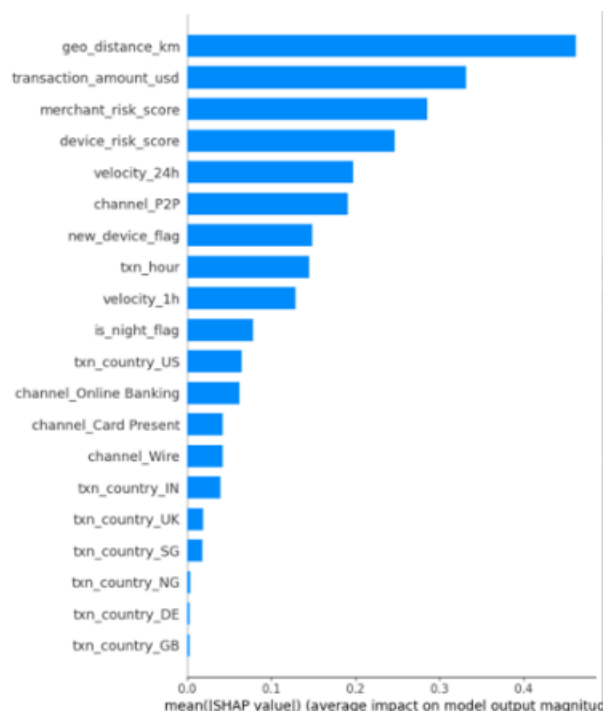


Figure 2. SHAP summary plot illustrating the relative contribution of key features to fraud predictions generated by the benchmark XGBoost model.

6. Operational Dashboard and Business Impact

6.1 Dashboard Overview

To complement the analytical framework, a suite of interactive Power BI dashboards was developed to transform model outputs into operational decision-support tools. Rather than presenting predictive results in isolation, the dashboards provide business users with a comprehensive view of fraud trends, operational risk, investigator priorities and key performance indicators.

The dashboard suite was designed for different stakeholder groups, ranging from senior management requiring executive oversight to operational fraud investigators responsible for reviewing and prioritising fraud alerts. This integration of analytics and business intelligence enables the framework to support both strategic reporting and day-to-day fraud operations.

6.2 Operational Insights

The dashboards present fraud intelligence across multiple operational perspectives, including transaction behaviour, customer characteristics, merchant activity and explainable fraud prioritisation. Interactive filtering enables investigators to explore fraud patterns by operational risk category, transaction channel, merchant category and customer segment.

The framework also incorporates operational features developed during the project, including risk categories, fraud reason codes, recommended investigator actions and SLA priorities. These enhancements provide additional business context that supports efficient alert triage and enables investigators to prioritise higher-risk transactions using transparent decision criteria.

Operational KPIs further support monitoring by providing visibility into fraud volumes, fraud rates, high-risk alerts and investigation priorities, allowing fraud teams to evaluate operational performance through a single analytical interface.



Figure 3. Power BI dashboards supporting executive oversight, fraud monitoring, alert prioritisation and operational performance management.

6.3 Business Value

The Explainable Fraud Intelligence Framework demonstrates how predictive analytics can be translated into practical operational value. By combining machine learning, explainability and business intelligence, the framework enables fraud investigators to move beyond simple fraud predictions and towards transparent, evidence-based decision-making.

From a business perspective, the framework offers several benefits, including improved fraud alert prioritisation, enhanced investigator confidence, greater transparency in model outputs and more effective allocation of investigative resources. Executive dashboards also provide management with timely operational insights that support monitoring, governance and continuous improvement of fraud detection processes.

Although developed within a simulated banking environment, the framework demonstrates how explainable fraud analytics can support modern fraud operations while maintaining human oversight and responsible AI principles.

7. Business Recommendations

The findings of this project demonstrate that effective fraud detection extends beyond predictive modelling and requires a balanced integration of analytics, explainability and operational decision support. Based on the outcomes of the Explainable Fraud Intelligence Framework, the following recommendations are proposed to enhance fraud management within financial institutions.

Strengthen Explainable Fraud Detection

Financial institutions should prioritise fraud detection models that provide both strong predictive capability and transparent decision-making. Explainable models improve investigator confidence, support governance requirements and enable fraud teams to understand the rationale behind high-risk alerts.

Adopt Risk-Based Alert Prioritisation

Fraud alerts should be prioritised according to operational risk rather than processed sequentially. Incorporating risk categories, reason codes, recommended investigator actions, and SLA priorities enables investigative resources to focus on transactions with the highest potential business impact.

Integrate Business Intelligence into Fraud Operations

Interactive dashboards should be embedded within operational workflows to provide real-time visibility of fraud trends, investigator workload and operational performance. Business intelligence supports faster decision-making while improving communication between operational teams and management.

Maintain Human Oversight

Machine learning should be viewed as a decision-support capability rather than a replacement for human expertise. Final fraud decisions should remain with trained investigators who can interpret analytical outputs alongside contextual business information.

Establish Continuous Model Governance

Fraud detection models should be monitored regularly to ensure continued performance as transaction behaviour and fraud patterns evolve. Periodic model validation, threshold reviews and

governance assessments will help maintain transparency, reliability and regulatory compliance throughout the model lifecycle.

8. Governance, Limitations and Future Enhancements

8.1 Responsible AI and Governance

Responsible AI formed an important consideration throughout the development of the Explainable Fraud Intelligence Framework. Model selection was based not only on predictive performance but also on transparency, interpretability and operational suitability. Selecting Logistic Regression as the operational model supports the investigator's understanding while improving accountability and governance within fraud management processes.

The framework also incorporates explainable operational outputs through risk categories, reason codes and recommended investigator actions, enabling fraud investigators to understand why transactions have been prioritised for review rather than relying solely on prediction scores.

8.2 Project Limitations

This project was developed using a synthetic banking dataset within a simulated operational environment. Consequently, the framework has not been validated using live production data or integrated with enterprise fraud management systems.

The project focuses on supervised fraud classification and does not incorporate real-time transaction monitoring, streaming analytics or automated case management. Furthermore, fraud patterns continually evolve, meaning that operational models require ongoing monitoring, recalibration and periodic retraining to maintain performance over time.

Despite these limitations, the project demonstrates how explainable machine learning can be successfully integrated with operational fraud intelligence to support transparent investigator-led decision-making.

8.3 Future Enhancements

Several opportunities exist to extend the framework beyond the current proof-of-concept. Future work could incorporate real-time transaction processing, behavioural analytics, graph-based fraud detection and advanced anomaly detection techniques to improve fraud identification.

Further enhancements may also include integration with enterprise case management platforms, automated model monitoring, continuous learning pipelines and deployment within a cloud-based operational environment. These developments would strengthen the framework's scalability while supporting more responsive and adaptive fraud operations.

9. Conclusion

This project successfully developed an Explainable Fraud Intelligence Framework that combines supervised machine learning, explainable artificial intelligence and business intelligence to support operational fraud detection within a simulated banking environment.

Rather than focusing solely on predictive accuracy, the framework demonstrates how fraud analytics can be integrated with explainability, operational risk prioritisation and interactive dashboards to improve transparency and support investigator-led decision-making. Comparative model evaluation identified Logistic Regression as the most appropriate operational model due to its balance of predictive capability, interpretability and operational suitability, while XGBoost provided valuable benchmarking and SHAP-based explainability.

The integration of operational risk categories, fraud reason codes, recommended investigator actions, SLA priorities and interactive Power BI dashboards extends the project beyond conventional fraud classification by delivering a practical decision-support framework for fraud operations.

Overall, the Explainable Fraud Intelligence Framework demonstrates how data science can be translated into operational business value through transparent analytics, responsible AI principles and effective business intelligence. Although implemented as a proof-of-concept, the framework provides a strong foundation for future development and illustrates how explainable analytics can support more effective, accountable and scalable fraud management practices.

Appendices

Appendix A – Key Dataset Variables

Variable	Description
transaction_id	Unique identifier for each transaction.
customer_id	Unique customer identifier.
merchant_id	Unique merchant identifier.
transaction_amount_usd	Monetary value of the transaction.
transaction_channel	Payment channel used for the transaction.
merchant_category	Merchant business category.
customer_segment	Customer segmentation used for analysis.
kyc_risk_band	Customer Know Your Customer (KYC) risk classification.
new_device_flag	Indicates whether the transaction originated from an unfamiliar device.
is_night_flag	Indicates whether the transaction occurred during higher-risk hours.
risk_category_v2	Operational fraud risk category (Low, Medium or High).
reason_codes	Explainable business reasons supporting fraud prioritisation.
recommended_action	Recommended investigator action generated by the framework.
sla_priority	Operational investigation priority assigned to the alert.
fraud_label	Target variable indicating fraudulent or legitimate transactions.

Appendix B – Model Performance Metrics

1. Cross-Validation Results

	Metric	Value
0	Mean ROC-AUC	0.612
1	Standard Deviation	0.035

Table B1. Five-fold stratified cross-validation results for the selected Logistic Regression model.

2. Threshold Evaluation

Operating Option	Threshold	Alerts Flagged	Precision	Recall	False Positives	False Negatives	False Positive Rate	Queue Hit Rate	Estimated Investigator Cases
Conservative	0.7	95	0.084	0.200	87	32	0.091	0.084	95
Balanced	0.5	263	0.080	0.525	242	19	0.252	0.080	263
Aggressive	0.3	547	0.049	0.675	520	13	0.542	0.049	547

Table B2. Operational comparison of alternative classification thresholds evaluated during model selection.

3. Logistic Regression Classification Report

Model	ROC-AUC	PR-AUC	Precision	Recall
Logistic Regression	0.628	0.076	0.08	0.525

Figure B1. Logistic Regression classification report.

4. XGBoost Classification Report

Classification Report:				
	precision	recall	f1-score	support
0	0.97	0.91	0.93	960
1	0.10	0.25	0.14	40
accuracy			0.88	1000
macro avg	0.53	0.58	0.54	1000
weighted avg	0.93	0.88	0.90	1000

Figure B2. XGBoost classification report.

Appendix C – Confusion Matrices

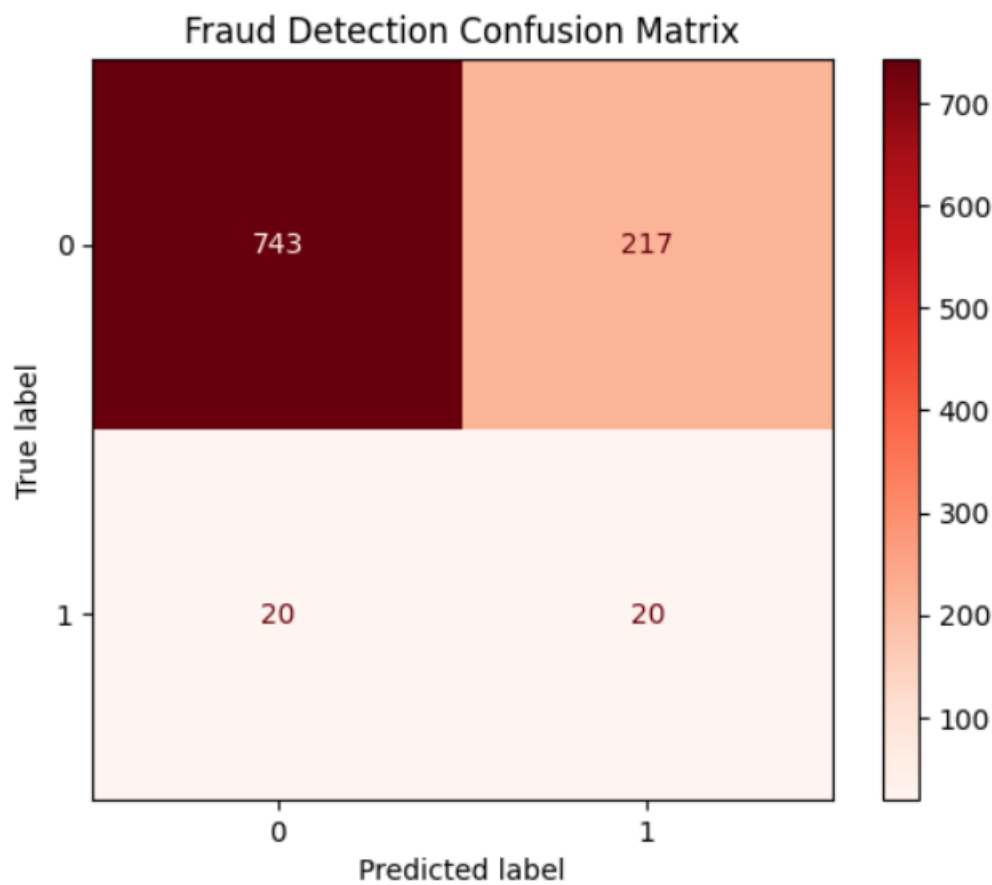


Figure C1. Confusion matrix for the selected Logistic Regression model.

Appendix D – Operational Model Card

Operational Model Card

Attribute	Description
Model Name	Explainable Fraud Intelligence Framework – Operational Fraud Model
Selected Operational Model	Logistic Regression
Benchmark Model	XGBoost (used for comparative evaluation and SHAP explainability)

Attribute	Description
Purpose	To support explainable fraud alert prioritisation and investigator-led decision support within a simulated banking environment.
Primary Inputs	Customer, merchant, transaction and behavioural intelligence features engineered from the integrated fraud intelligence dataset.
Primary Outputs	Fraud probability score, operational risk category, reason codes, recommended investigator actions and SLA priority.
Primary Users	Fraud Analysts, Fraud Investigators, Risk & Compliance Teams, and Operational Management.
Decision Role	Decision-support tool only. Final fraud decisions remain subject to human investigator review.
Governance Controls	Data leakage prevention, threshold governance, human-in-the-loop review, operational KPI monitoring and periodic model validation.
Key Limitation	Developed using a synthetic dataset as a proof-of-concept and not validated within a live production fraud environment.